



SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL
SCIENCES



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE NAME: Theory of Computation

COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO2: Design Finite Automata DFA, NFA, ϵ -NFA and demonstrate their equivalence for recognizing regular languages. (BL:3)

Assessment Tool Weightage: 20%

QUESTIONS: 5
Duration : 1 Hour

Total Marks:50

Scenario Based

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



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1. A company is developing a smart security system for a research laboratory. Every employee entering the lab uses a digital access card that generates a sequence of symbols:

- $a \rightarrow$ Valid ID scan
- $b \rightarrow$ Biometric verification
- $c \rightarrow$ Door access confirmation

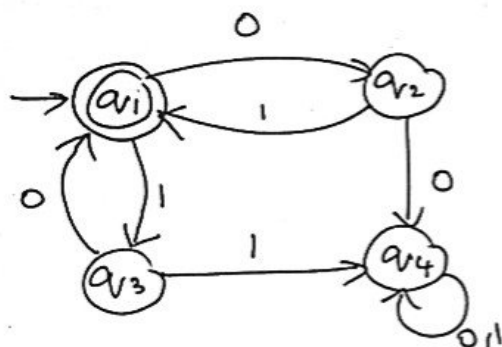
The security rule states that:

1. Every valid access sequence must start with a .
2. It may contain any number of biometric checks (b).
3. The sequence must end with c .

Examples of valid sequences: $ac, abc, abbc, abbbc$

2. Given a DFA used for validating a specific PIN pattern $bbbc, bca, c, ca, caaaaa$ minimize the DFA and explain how state reduction improves performance and efficiency in embedded systems

3. Demonstrate how a given finite automaton can be converted into a regular expression



4. Consider the following grammar

$S \rightarrow aSc | T | U | e$

$T \rightarrow bTc | e$

$U \rightarrow aUb | e$

A) What is the language defined by the grammar?

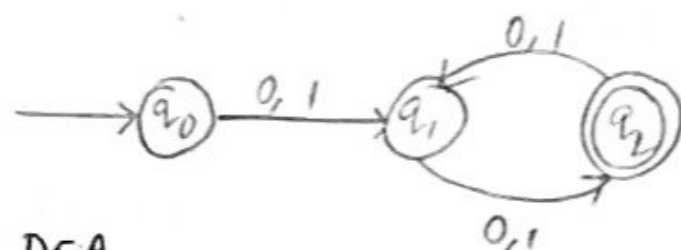
B) Give the derivation tree for the word $aaabbc$

C Grammar is Ambiguous if there exists two distinct derivations for a word. Is the above grammar ambiguous? Justify.

5. i) Show that if L_1 and L_2 are regular then $(L_1 + L_2)$ also regular

ii) Is the language $\{a^p \mid p \text{ is prime}\}$ regular? Justify

Q₁ Design the Regular Expression for the Given automata.



Given DFA

state	0	1	
$\rightarrow q_0$	q_1	q_1	$q_0 \xrightarrow{0} q_1, q_0 \xrightarrow{1} q_2$
q_1	q_2	q_2	$q_1 \xrightarrow{0,1} q_2$
q_2	q_1	q_1	$q_2 \xrightarrow{0,1} q_1$

$\therefore$ only strings having even length

Regular expression $\rightarrow$ one pair of symbols is

$(0+1)(0+1)$ additional pairs are allowed.

$((0+1)(0+1))^+$

$\therefore$ DFA accepts all binary strings of even length.

Q₂ To determine the language generated by the grammar.

Given, $S \rightarrow aSc | T | U | \epsilon$

$T \rightarrow bTc | \epsilon$

$U \rightarrow aUb | \epsilon$

Rule 1

$$S \rightarrow aSc$$

Each recursion adds

one a

one c

Generated strings

ϵ

ac

aacc

aaaccc

$$L_1 = \{a^n c^n \mid n \geq 0\}$$

Rule 3

$$U \rightarrow aUb$$

Generated strings

ϵ

ab

aabb

aaabbb

$$L = L_1 \cup L_2 \cup L_3$$

$$L_3 = \{a^n b^n \mid n \geq 0\}$$

8) derivation tree for aaabbbc

Leftmost derivation

S

$$\Rightarrow aSc$$

$$\Rightarrow aUc$$

$$\Rightarrow aaUbc$$

$$\Rightarrow aa a U b b c$$

$$\Rightarrow aa a \epsilon b b c$$

$$\Rightarrow aa a b b c$$



c) Ambiguity

a grammar is ambiguous if one string has two different parse trees

For $aaabbc \rightarrow$ only one derivation exists

$\therefore$ The grammar is unambiguous.

Q3

Regular Expression

$$\boxed{00^*1 + 11^*0}$$

to design an ϵ -NFA for the Regular expression

$$00^*1 + 11^*0$$

Expression 1

00^*1

Accepted strings $\rightarrow 01$

001

0001

00001

Expression 2

11^*0

Accepted strings:-

10

110

1110

11110

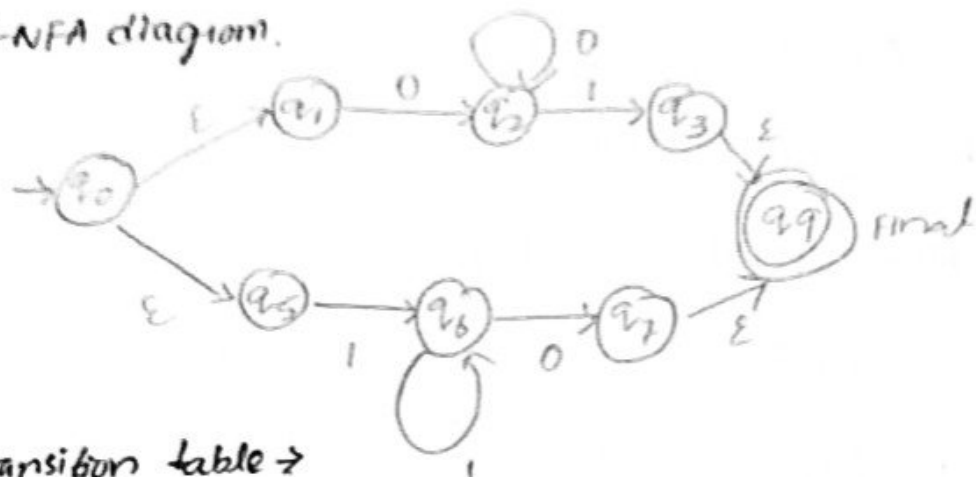
Thompson construction

q_0 = start state

q_9 = final state.

State	Input	Next state
q_0	ϵ	q_1
q_0	ϵ	q_5
q_1	0	q_2
q_2	0	q_2
q_3	1	q_3
q_4	ϵ	q_9
q_5	1	q_6
q_6	0	q_7
q_7	ϵ	q_9

ϵ -NFA diagram.



Transition table $\rightarrow$

State	ϵ	0
q_0	q_1, q_5	$\emptyset$
q_1	$\emptyset$	q_2
q_2	$\emptyset$	q_2
q_3	q_4	$\emptyset$
q_5	$\emptyset$	q_6
q_6	$\emptyset$	q_6
q_7	q_4	$\emptyset$
q_4	$\emptyset$	$\emptyset$

$\therefore \epsilon$ -NFA correctly accepts all strings represented by.

$$00^*1 + 11^*0$$

Qy (b) no show $L_1 + L_2$ is also Regular.

A language is regular if it is accepted by a finite automation.

Suppose L_1 is accepted by DFA

$$M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$$

and L_2 is accepted by DFA

$$M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$$

Construct a new DFA

$$M = (Q, \Sigma, \delta, q, F)$$

States

start
state

Transition.

$$Q = Q_1 \times Q_2$$

$$(q_1, q_2)$$

$$\delta((p, r), a) = (\delta_1(p, a), \delta_2(r, a))$$

Final states $\rightarrow$

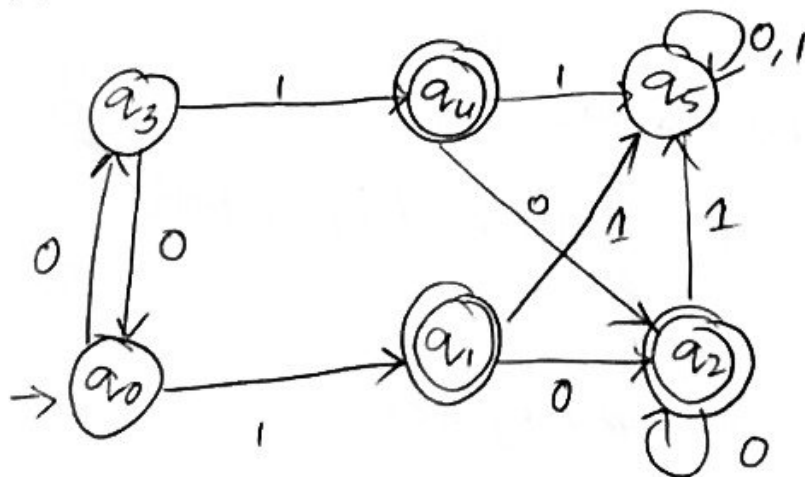
$$F = (F_1 \times Q_2) \cup (Q_1 \times F_2)$$

hence,

$$L(M) = L_1 \cup L_2$$

$L_1 + L_2$ is Regular

Minimization of DFA



by using partitioning algorithm.

final $\rightarrow q_1, q_2, q_4$

non-final $\rightarrow q_0, q_3, q_5$

$$P_0 \Rightarrow \left\{ \underbrace{\{q_1, q_2, q_4\}}_{\text{Block 1}}, \underbrace{\{q_0, q_3, q_5\}}_{\text{Block 2}} \right\}$$

$P_1 \Rightarrow$

from block 1 $\Rightarrow \{q_1, q_2, q_4\}$

states	$\delta(0)$	$\delta(1)$
q_1	q_2	q_5
q_2	q_2	q_5
q_4	q_2	q_5

Same.

$q_1, q_2, q_4 \Rightarrow \{q_1, q_2, q_4\} \rightarrow \text{equivalent}$

Block 2 $\Rightarrow \{q_0, q_3, q_5\}$

states	$\delta(0)$	$\delta(1)$
q_0	q_3	q_1
q_3	q_0	q_4
q_5	q_5	q_5

compare = $\{q_0, q_3\}$

$\Rightarrow \delta_0$

$q_0 \rightarrow q_3$
 $q_3 \rightarrow q_0$ } $\rightarrow$ Final

$\delta(1)$

$q_0 \rightarrow (q_1)$
 $q_3 \rightarrow (q_2)$ } $\rightarrow$ non Final

$q_0 \neq q_3$

compare = $\{q_0, q_5\}$

$\delta(0)$

$q_0 \rightarrow q_3$
 $q_5 \rightarrow q_5$ } $\rightarrow$ non Final

$\delta(1)$

$q_0 \rightarrow (q_1)$
 $q_5 \rightarrow q_5$ } $\rightarrow$ non Final

$q_0 \neq q_5$

compare = $\{q_3, q_5\}$

$\delta(0)$

$q_3 \rightarrow q_0$
 $q_5 \rightarrow q_5$ } $\rightarrow$ non Final

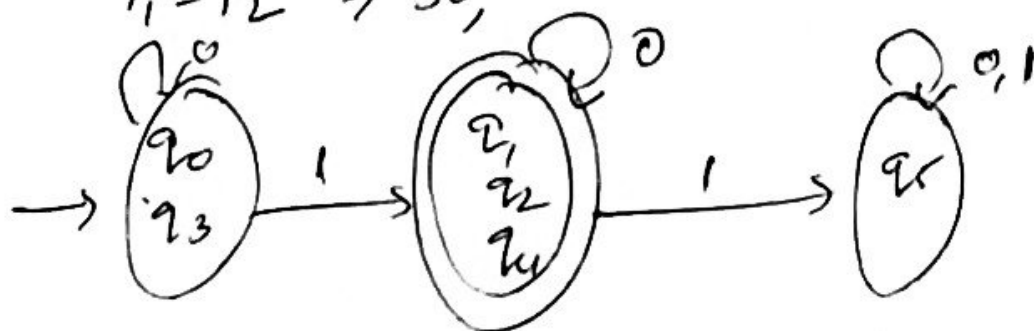
$\delta(1)$

$q_3 \rightarrow (q_4) \rightarrow$ Final
 $q_5 \rightarrow q_5 \rightarrow$ non

$q_3 \neq q_5$

$P_1 \Rightarrow \{\{q_1, q_2, q_4\}, \{q_0, q_3\}, \{q_5\}\}$

$P_1 = P_2 \Rightarrow \delta_0,$





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RUBRICS FOR CALCULATION

Criteria	Weight age	Q1 (10)	Q2		Q4 (10)	Q5 (10)	Total Marks (50)
Conceptual Understanding	30	3	100		3	3	15
Accuracy of Answer	25	2.5	2.5	2.5	2.5	2.5	12.5
Application of Knowledge	20	2	2	2	2	2	10
Completeness of Response	15	1.5	1.5	1.5	1.5	1.5	7.5
Clarity & Presentation	10	1	1	1	1	1	5
Total per Question	10 Marks	/10	/10	/10	/10	/10	/50